Quiz#10

Section: _____ **Group:** _____

Name(print): _____ **Student ID:** _____

A uniform rod of mass *M* and length *L* is shown in the diagram, with an axis of rotation located a distance *L*/3 from one end. Two forces of equal magnitude *F* are exerted on the rod: one at the left end, perpendicular to the rod (90° angle), and the other at a point *L*/2 from the axis of rotation, forming a 110° angle with the rod. Clockwise is defined to be the positive rotational direction.

Using the given variables *M*, *L*, *F*, and appropriate trigonometric functions, determine expressions for the following:

a. The moment of inertia of the rod about the axis of rotation.

b. The net torque about the axis of rotation.

c. The angular acceleration of the rod.

